Functional Forecasting of Turkish Electricity Load Curves - Preliminary Report

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Object and functional representation

Let $Y_d(h)$ denote Turkish electricity load on day d, hour $h = 0, \dots, 23$. The empirical object is the daily load curve

$$Y_d = \{Y_d(h) : h = 0, \dots, 23\}.$$

The cleaned data are stored as daily curve matrices

$$Y = [Y_d(h)]_{d,h}, \qquad \widehat{Y}^{off} = [\widehat{Y}_d^{off}(h)]_{d,h}, \qquad E = [Y_d(h) - \widehat{Y}_d^{off}(h)]_{d,h},$$

with 24 hourly observations per retained day.
Each curve is represented by a B-spline basis,

$$Y_d(h) \simeq \sum_{m=1}^{M} c_{dm} B_m(h), \qquad M = 12.$$

FPCA gives

$$Y_d(h) = \mu(h) + \sum_{k=1}^{K} \xi_{dk} \phi_k(h) + e_d(h), \qquad K = 4.$$

Let

$$\xi_d = (\xi_{d1}, \dots, \xi_{dK})', \qquad \Phi(h) = (\phi_1(h), \dots, \phi_K(h))'.$$

Then

$$Y_d(h) = \mu(h) + \Phi(h)'\xi_d + e_d(h), \qquad \widehat{Y}_d(h) = \widehat{\mu}(h) + \widehat{\Phi}(h)'\widehat{\xi}_d.$$

All forecasts are rolling-origin forecasts: for target day d, estimation uses only information dated $< d$. The evaluation period starts on 2023-01-01 and contains 26,304 hourly observations.

Forecasting specifications

Benchmarks:

$$\widehat{Y}_d^{SN}(h) = Y_{d-7}(h), \qquad \widehat{Y}_d^{OFF}(h).$$

Separate weekly score autoregression:

$$\xi_{dk} = \alpha_k + \beta_k \xi_{d-7,k} + u_{dk}, \qquad k = 1, \dots, K.$$

Weekly FPCA VAR:

$$\xi_d = a + A_7 \xi_{d-7} + u_d, \qquad A_7 \in \mathcal{R}^{K \times K}.$$

FPCA VAR(1,7):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d, \qquad A_1, A_7 \in \mathcal{R}^{K \times K}.$$

Augmented score model:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d,$$

where z_d is predetermined at date d.
Final corrected forecast:

$$\widehat{Y}_d^{final}(h) = \widehat{\mu}(h) + \widehat{\Phi}(h)'\widehat{\xi}_d + \lambda_d \widehat{r}_d(h).$$

Predictor construction

The largest predictor vector is

$$z_d = \left(z_d^{cal'}, z_d^{hol'}, z_d^{shape'}, z_d^{deriv'}\right)'.$$

Ramp and early-load variables:

$$R_{d-1}^{m} = Y_{d-1}(9) - Y_{d-1}(6), \qquad R_{d-7}^{m} = Y_{d-7}(9) - Y_{d-7}(6),$$

$$L^{e}_{d-1} = \frac{1}{3}\sum_{h\in\{5,6,7\}} Y_{d-1}(h).$$

Calendar controls:

$$z^{cal}_d = \Big(weekend_d, \sin\frac{2\pi doy_d}{365.25}, \cos\frac{2\pi doy_d}{365.25}\Big)'.$$

Holiday controls include fixed holidays, religious holidays, holiday eve, post-holiday, near-holiday, and bridge-day indicators:

$$Bridge_d = 1\{dow_d \in \{Mon, Fri\}\}1\{d \notin \mathcal{H}\}1\{d-1 \in \mathcal{H} \text{ or } d+1 \in \mathcal{H}\}.$$

Lagged load-shape variables:

$$\bar{Y}_{d-1} = \frac{1}{24}\sum_{h=0}^{23} Y_{d-1}(h), \qquad P_{d-1} = \max_h Y_{d-1}(h), \qquad M_{d-1} = \min_h Y_{d-1}(h),$$

$$G_{d-1} = P_{d-1} - M_{d-1}, \qquad \bar{Y}^{(q)}_d = \frac{1}{q}\sum_{j=1}^{q}\left(\frac{1}{24}\sum_{h=0}^{23} Y_{d-j}(h)\right), \quad q \in \{7, 28\}.$$

Derivative features are computed from the smoothed curve,

$$DY_d(h) = \frac{d}{dh} Y_d(h),$$

and summarized by

$$\overline{DY}_{d-1}, \quad SD(DY_{d-1}), \quad \max_h DY_{d-1}(h), \quad \min_h DY_{d-1}(h), \quad \max_h DY_{d-1}(h) - \min_h DY_{d-1}(h).$$

Targeted intraday changes:

$$R^{e}_{d-1} = Y_{d-1}(7) - Y_{d-1}(5), \qquad R^{ev}_{d-1} = Y_{d-1}(20) - Y_{d-1}(17), \qquad R^{end}_{d-1} = Y_{d-1}(23) - Y_{d-1}(20).$$

Adaptive correction uses past residuals

$$\widehat{e}_s(h) = Y_s(h) - \widehat{Y}^{base}_s(h), \qquad s < d,$$

and applies

$$\widehat{Y}^{corr}_d(h) = \widehat{Y}^{base}_d(h) + \lambda_d \widehat{r}_d(h),$$

where $\widehat{r}_d(h)$ is estimated from recent residual patterns and λ_d is selected on a validation window.

Evaluation

For model m,

$$e^{(m)}_{dh} = Y_d(h) - \widehat{Y}^{(m)}_d(h).$$

$$MAE_m = \frac{1}{N}\sum_{d,h} |e^{(m)}_{dh}|, \qquad RMSE_m = \left[\frac{1}{N}\sum_{d,h}(e^{(m)}_{dh})^2\right]^{1/2}, \qquad Bias_m = \frac{1}{N}\sum_{d,h} e^{(m)}_{dh}.$$

$$Gain^{MAE}_m = 100\frac{MAE_{OFF} - MAE_m}{MAE_{OFF}}, \qquad Gain^{RMSE}_m = 100\frac{RMSE_{OFF} - RMSE_m}{RMSE_{OFF}}.$$

Current results

Model	MAE	RMSE	Bias	MAE gain	RMSE gain
Seasonal naive Y_{d-7}	1912.46	3129.98	39.85	-57.60%	-79.85%
Official forecast	1213.50	1740.30	466.87	0.00%	0.00%
FPCA VAR(1,7) scores	1071.13	1562.67	-55.00	11.73%	10.21%
FPCA VAR(1,7) + ramp/calendar	1031.79	1493.24	112.34	14.97%	14.20%
FPCA VAR(1,7) + derivative features	890.01	1286.78	91.28	26.66%	26.06%
FPCA VAR(1,7) + holiday/load-shape + adaptive correction	666.49	996.22	66.46	45.08%	42.76%
FPCA VAR(1,7) + holiday/load-shape/derivative + adaptive correction	**640.10**	**967.82**	**103.39**	**47.25%**	**44.39%**

The final model improves on the official forecast by 47.25% in MAE and 44.39% in RMSE. Relative to the weekly seasonal naive benchmark,

$$100\frac{1912.46 - 640.10}{1912.46} = 66.53\%.$$